

Tyler Kuipers

SOFTWARE ARCHITECT

Edmonton, Canada

☎ (587) 879-4626 | ✉ tkuipers123@gmail.com | 🌐 tkuipers.ca | 📱 [tkuipers](#) | 🌐 [tyler-kuipers](#)

About Me

Software Architect with 10+ years of experience designing and implementing scalable microservices, APIs, and distributed systems. Proficient in TypeScript, JavaScript, and multiple backend languages. Proven track record leading cross-team technical initiatives, architecting cloud infrastructure serving thousands of users, and supporting production systems. Focus on test-driven development, API design, and establishing technical standards across development teams.

Work Experience

Alberta Motor Association

INSURANCE SOFTWARE ARCHITECT

Edmonton, Alberta

Jan 2023 - Present

- Led cross-team initiatives architecting cloud-based microservices for pay-as-you-go insurance product; supported 1400+ policies in first 4 months with scalable, distributed architecture. (AWS, Terraform, Java)
- Designed and implemented RESTful APIs using OpenAPI Specification (OAS) powering online auto sales platform, ensuring performance and reliability under high-traffic loads. (OpenAPI, TypeScript, Gosu, Playwright)
- Modernized legacy monolithic system into scalable microservices architecture with improved maintainability. (AWS, Terraform, Kubernetes)
- Built DevOps automation tooling that reduced environment provisioning time by 90%; supported production systems through monitoring and debugging. (AWS, Terraform, Python, Jenkins, Docker)
- Designed technical architecture for multi-team integration with chatbot solution enabling automated phone payments. (Java, OpenAPI)
- Architected overhaul of online payment system to support omnichannel experience across web and mobile platforms.
- Established coding standards and API design best practices across 12-person development team; provided ongoing technical guidance on architecture decisions and code quality.

Vinted

SOFTWARE DEVELOPER

Berlin, Germany

Sept 2022 - Dec 2022

- Contributed to architecture design and full-stack implementation of new shipping platform supporting multi-carrier integration. (Ruby on Rails, Shopify, Kubernetes)
- Developed and maintained automated test suite ensuring reliability and performance across carrier integrations.

Arterys Inc. (Acquired by Tempus Labs)

SOFTWARE DEVELOPER

Calgary, Alberta (Remote)

Aug 2021 - Aug 2022

- Integrated multiple medical detection AI models into SaaS platform, enabling automated diagnostic workflows. (TypeScript, Node.js)
- Improved performance by reducing AI algorithm turnaround time through infrastructure optimization and caching strategies. (AWS, Chef, Redis)
- Supported production services by monitoring system health, logs, and observability metrics to proactively identify issues before client impact. (Elasticsearch, FullStory, Kibana)
- Delivered and maintained full-stack product enhancements; collaborated using Git and GitHub workflows for code review and version control.

Alberta Motor Association

SOFTWARE DEVELOPER | SOFTWARE TEAM LEAD

Edmonton, Alberta

Jun 2018 - Feb 2021 | Feb 2021 - Aug 2021

- Led 6 person development team implementing PolicyCenter platform and contributed to organization-wide architectural standards initiative. (Gosu)
- Designed and built automated test framework using test-driven development that significantly reduced bug rates and improved code quality across subsequent projects.
- Architected and implemented online payment and booking solutions integrating third-party APIs with Guidewire platform, enabling customer self-service scheduling. (Gosu, JavaScript, OAuth)
- Conducted technical assessment for Guidewire cloud migration, evaluating architecture modernization scope and effort.
- Served as Scrum Master, improving team efficiency and facilitating developer-driven process improvements. (Azure DevOps, Agile)

Pleasant Solutions

SOFTWARE DEVELOPER

Edmonton, Alberta

Jan 2018 - Jun 2018

- Designed and developed Human-Machine Interface (HMI) for concrete distribution plant using test-driven development methodology. (C#, TDD)
- Led full-stack technical implementation on client-customized software projects.

Lockheed Martin Canada

SOFTWARE ENGINEER ASSOCIATE

Halifax, Nova Scotia

Jan 2017 - Jan 2018

- Designed and implemented Data Fusion module for Combat Management Systems research project, integrating multiple data sources. (Java, XML, SQL, Oracle ESB)
- Developed full-stack software modules for underwater sensor management and status reporting using Swing framework. (Java, Swing)

The King's University

SOFTWARE DEVELOPER/RESEARCH ASSISTANT

Edmonton, Alberta

May 2014 - Sep 2016

- Designed and built full-stack web application enabling spatial and temporal data visualization for Edmonton River Atlas project. (PHP, HTML, JavaScript, PostgreSQL, PostGIS)
- Authored undergraduate thesis documenting dynamic spatial-temporal referencing methodology and system architecture. (L^AT_EX)

Education

The Open University

MSC. COMPUTING (SOFTWARE ENGINEERING)

Currently pursuing Master's in Computing at The Open University.

Milton Keynes, England (Remote)

May 2024 - Ongoing

The King's University

B.S. IN COMPUTER SCIENCE

Edmonton, Alberta

Sep 2012 - Dec 2016

Technologies

Languages

TypeScript, JavaScript, Java, Gosu, Python, Ruby, C#, SQL, PHP

Frameworks & Libraries

Node.js, Ruby on Rails, Spring, Swing, React, Playwright

Databases & Caching

PostgreSQL, PostGIS, Redis, Oracle, SQL Server

Cloud & Infrastructure

AWS (EC2, Lambda, RDS, S3), Terraform, Kubernetes, Docker, Chef

DevOps & CI/CD

Jenkins, GitHub Actions, Azure DevOps, Git, GitHub, Test-Driven Development (TDD)

APIs & Integration

REST, OpenAPI (OAS), OAuth, Shopify, Guidewire, Oracle ESB

Observability & Monitoring

Elasticsearch, Kibana, FullStory, CloudWatch

Methodologies

Agile, Scrum, Automated Testing, Microservices Architecture, Distributed Systems